WORKED SOLUTION

STEP II 2008, Question 9

Mechanics • Projectile Motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

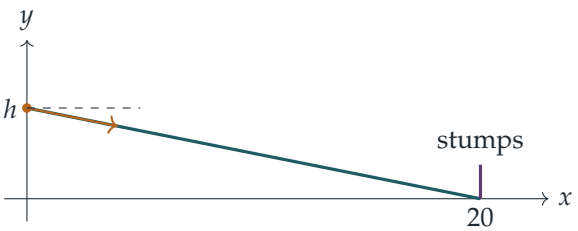
step.maths.org/questions/08-s2-q9

The key idea

The horizontal motion fixes the time of flight in terms of α, and putting that time into the vertical motion gives one equation linking α and h. Writing $\sec^2\alpha = 1 + \tan^2\alpha$ turns it into a quadratic in $\tan\alpha$, the standard trick for "which angle hits the target?" problems. After that, common sense about cricket picks the right root, and small-angle approximations finish the job.

Solution

We model the ball as a particle moving freely under gravity, with no air resistance and no spin. Take the origin on the ground directly below the point of projection, with x horizontal towards the stumps and y vertically up, so the ball starts at $(0, h)$ with speed $40\,\mathrm{m\,s^{-1}}$ at angle α above the horizontal (a negative α means it is aimed downwards).



Part (i)

Horizontally there is no force, so $x = 40t\cos\alpha$. The ball reaches $x = 20$ when

$$t = \frac{20}{40\cos\alpha} = \frac{1}{2\cos\alpha}.$$

Vertically the acceleration is $-10\,\mathrm{m\,s^{-2}}$, so

$$y = h + 40t\sin\alpha - 5t^2.$$

The ball is at ground level, $y = 0$, at this time, so

$$0 = h + 40\sin\alpha \cdot \frac{1}{2\cos\alpha} - \frac{5}{4\cos^2\alpha} = h + 20\tan\alpha - \tfrac{5}{4}\sec^2\alpha.$$

Now we use $\sec^2\alpha = 1 + \tan^2\alpha$ and write $T = \tan\alpha$:

$$h + 20T - \tfrac{5}{4}(1 + T^2) = 0 \quad \Longleftrightarrow \quad 5T^2 - 80T + 5 - 4h = 0.$$

By the quadratic formula,

$$T = \frac{80 \pm \sqrt{6400 - 20(5 - 4h)}}{10} = \frac{80 \pm \sqrt{6300 + 80h}}{10},$$

so the two possible values are

$$\tan\alpha = 8 \pm \sqrt{63 + \tfrac{4}{5}h}.$$

Both are real, since $h \geq 0$.

Ask yourself Which root would a fast bowler actually use?

Since $\sqrt{63} \approx 7.9$, the larger root gives $\tan\alpha \approx 16$, so $\alpha \approx 86^\circ$: a towering lob that would rise about 80 m and take around 8 s to come down on the stumps. That is no fast bowler's delivery. The smaller root,

$$\tan\alpha = 8 - \sqrt{63 + \tfrac{4}{5}h},$$

is the appropriate one. For a realistic release height (a couple of metres), $63 + \frac{4}{5}h$ is close to 64, so $\tan\alpha$ is close to 0. For example, any h between 0 and 3 gives $|\tan\alpha| < 0.09$. So α is a small angle and $\cos\alpha \approx 1$, and the time to reach the stumps is

$$t = \frac{1}{2\cos\alpha} \approx \tfrac{1}{2}\text{ s},$$

as required. (In fact $t = \frac{1}{2}\sqrt{1 + \tan^2\alpha}$, which is within 0.5% of $\frac{1}{2}$ whenever $|\tan\alpha| < 0.1$.)

Part (ii)

The ball is projected below the horizontal exactly when $\tan\alpha < 0$. The larger root is always positive, so we use the smaller one:

$$8 - \sqrt{63 + \tfrac{4}{5}h} < 0 \iff \sqrt{63 + \tfrac{4}{5}h} > 8 \iff 63 + \tfrac{4}{5}h > 64 \iff h > \tfrac{5}{4},$$

where squaring is reversible because both sides are positive. So the bowler projects the ball below the horizontal when $h > 1.25$. When $h = 1.25$ exactly, the ball is projected horizontally.

Part (iii)

When $h = 2.5$ we know from part (ii) that α is small and negative. With $\cos\alpha \approx 1$ and $\sin\alpha \approx \alpha$ (in radians), we have $\tan\alpha \approx \alpha$ and $\sec^2\alpha \approx 1$, so the equation $0 = h + 20\tan\alpha - \frac{5}{4}\sec^2\alpha$ from part (i) becomes

$$0 \approx 2.5 + 20\alpha - 1.25, \qquad \text{so} \qquad \alpha \approx -\frac{1.25}{20} = -0.0625 \text{ rad}.$$

Converting to degrees,

$$\alpha \approx -0.0625 \times \frac{180}{\pi} \approx -3.58^\circ,$$

so $\alpha \approx -3.6^\circ$. The ball is released about 3.6° below the horizontal.

Another way to see it: $\tan\alpha = 8 - \sqrt{65} = 8 - 8\sqrt{1 + \frac{1}{64}} \approx 8 - 8(1 + \frac{1}{128}) = -\frac{1}{16}$, using the binomial approximation. This gives the same answer.

Checks

- *Exact value.* $\tan\alpha = 8 - \sqrt{65} = -0.06226$, so $\alpha = -3.563^\circ$, which is -3.6° to two significant figures, in agreement with the approximation.
- *Simulation.* Stepping the motion numerically from $(0, 2.5)$ at $40\,\text{m}\,\text{s}^{-1}$ and -3.563°, the ball reaches $x = 20$ at height $0.0000\,\text{m}$ after $0.501\,\text{s}$.
- *Boundary.* At $h = 1.25$ the smaller root is $8 - \sqrt{64} = 0$: the ball is projected horizontally, falls $5(0.5)^2 = 1.25\,\text{m}$ in the half-second flight, and lands exactly at the stumps.

Common slips

- Forgetting $\sec^2\alpha = 1 + \tan^2\alpha$ and trying to solve in $\sin\alpha$ and $\cos\alpha$ separately, which gets messy.
- Giving the angle in radians, or using the small-angle approximation in degrees: $\sin\theta \approx \theta$ only holds with θ in radians.
- Dropping the minus sign in part (iii). The angle is below the horizontal, as part (ii) predicts.

Beyond STEP

The two-root structure is the general fact that a projectile with a fixed speed reaches any point inside its "envelope" (a parabola of safety) by two trajectories. On the envelope the two merge into one, and points outside it cannot be reached at all. Real cricket balls are also affected by drag and by the Magnus effect from spin, which leads to nonlinear differential equations that usually have to be solved numerically. Kleppner

and Kolenkow, *An Introduction to Mechanics*, treats projectiles and resisted motion at first-year level.

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